

TED(10)- 4087

Reg.No.

(REVISION – 2010)

Signature

MODEL QUESTION PAPER

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

ADVANCED PROCESS CONTROL

(Common for EI and IT)

[Time : 3 hours]

(Maximum Marks : 100)

PART – A

I Answer the following questions in one or two sentences. Each question carries 2 marks

1. What are the disadvantages of Cascade control system?
2. Mention any two application of ratio control system?
3. What is DCS?
4. List two advantages of PLC
5. Draw any three P&ID signal and process lines (5x2 = 10)

PART – B

II Answer any five of the following questions. Each question carries 6 marks.

1. Write the characteristics of Continuous process
2. What is Cascade control system? Write some applications of it.
3. Explain the features of DCS.
4. Describe relay logic control with an example
5. List some disadvantages of relay logic control
6. Describe Ziegler – Nichols methods of process control loop tuning
7. Describe general features of SCADA (5x6 = 30)

PART - C

(Answer one full question from each unit. Each question carries 15 marks.)

MODULE -1

- III 1. Write short notes on

(a) Ratio control system (b) Split range control system and (c) feedback control system 15

OR

- III 1. With suitable example explain the characteristics of
- (a) Multivariable process control and
- (b) Compound variable process control. 8
2. Explain feedback control system and list two applications. 7

MODULE -II

- V 1. Explain Data acquisition system with suitable diagram 7
2. Explain the general features of Centralized Computer control. 8

OR

- VI 1. Draw the block diagram Data Logging System and explain 9
2. Write advantages and disadvantages of supervisory control 6

MODULE -III

- VII 1. Explain the operation of PLC 7
2. write short notes on instruction sets used in PLC 8
- OR
- VIII 1. Draw the block diagram of PLC and explain 8
2. What is Relay Logic Control? Explain with an example 7

MODULE – IV

- IX 1. Draw the block diagram of SCADA and explain 9
- 2 Describe the importance of Process and Instrument diagrams 6

OR

- X 1. Write advantages and disadvantages of SCADA . 6
- 2 Draw & Explain P&I diagrams of Heat Exchanger. 9
